Statistical and Deep Learning Approaches for Mapping 3-D Ground Ice Properties

by

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Abstract

Ground ice is a crucial component of permafrost, and mapping its distribution and properties is critical for understanding the implications of climate change on local environments. This project investigates creating a 3D ground ice map of Canada based on borehole data in the Inuvik-Tuktoyaktuk highway region. The research aims to develop methods to integrate borehole vertical profile point-based data with land surface spatial raster data, such as satellite remote sensing and geomorphometry, by investigating the correlations between surface and subsurface characteristics, and developing predictive models for ground ice parameters.

Keywords: ground ice; deep learning; data fusion; feature-wise linear modulation; permafrost; remote sensing

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Chapter 1

Introduction

Ground ice, a key component of the permafrost ecosystem, is known to play an essential role in determining the thermal and mechanical properties of permafrost, as well as the vulnerability of the latter to climate change. When ground ice thaws, it can lead to land subsidence and soil erosion, which in turn can have severe impacts on the environment, infrastructure, and local communities.

Given the crucial role of ground ice in the permafrost ecosystem, there is a growing need for predictive models of ground ice distribution and properties. In recent years, advances in machine learning and data science have enabled researchers to build increasingly sophisticated models that can handle complex, multi-modal data and incorporate field knowledge. In particular, deep learning has shown promising results due to its ability to learn complex patterns from diverse datasets.

1.1 Related Work

1.1.1 Ground Ice

While existing studies have focused on modeling ground ice in a 2D context, our objective is to extend this modeling approach into the depth dimension. Relevant references from soil science literature support our pursuit of this multidimensional understanding of ground ice behavior.

The state-of-the-art ground ice and permafrost map of Canada [3] incorporates paleogeography in an expert-system approach to predict the distribution of relict ice, segregated ice, and wedge ice in northern Canada. The predictions are based on a range of environmental factors, showing improved spatial accuracy compared to previous mappings. However, the GIMC paper also emphasizes the need for further field observations and volumetric ice estimates for comprehensive validation. This work serves as a valuable reference for our ultimate goal in 3D ground ice investigations. The aim is to build upon this foundation, transitioning from the 2D mapping presented in [3] to a 3D ground ice map.

The paper "Probabilistic Representation of Genetic Soil Horizons" [4] introduces a novel approach using machine learning to represent soil horizons probabilistically, advancing understanding of soil horizon variability. Simultaneously, the study "Statistical and Geostatistical Analysis of Permafrost Extent: Mackenzie Valley, Canada" [5] analyzes permafrost in the Mackenzie Valley using statistical and geostatistical methods, forming the basis for our work, which extends the analysis to three dimensions and incorporates diverse data modalities for a comprehensive understanding of ground ice behavior in Canada.

1.1.2 Deep Learning

In recent years, employing deep learning techniques on land raster data has become a prominent avenue for advancing geospatial analyses. Residual networks (ResNets) [6] and multi-layer perceptrons (MLPs) [7] have emerged as powerful architectures for extracting intricate patterns and features from complex spatial datasets, as demonstrated in [8, 9]. ResNets, with their unique residual connections, enable the training of deep networks more effectively, mitigating issues like vanishing gradients. On the other hand, MLPs provide a versatile framework for learning complex relationships within raster data. Additionally, the integration of data fusion techniques for handling multi-modal information has been pivotal [10]. This involves combining diverse datasets, such as tabular borehole data and land surface raster data, within a unified deep learning framework. These advances underscore the evolving applications of deep learning to land raster data, promising richer insights into the intricate interplay of environmental factors.

1.2 Thesis Outline

This project aims to explore a deep learning-based predictive model for ground ice characteristics in the Arctic. To achieve this, the project leverages two main data modalities: tabular borehole data and land surface raster data. The borehole data contains information such as visible ice content and material type, while the land surface raster data contains information such as topographic and geomorphic variables, roughness, and seasonal heave/subsidence activity. The FiLM architecture [11] provides a solution to the challenge of incorporating multiple types of data into a single deep learning model.

The key contributions of this thesis consist of establishing the potential of mapping sub-surface ground ice characteristics through the following points:

- A rigorous data fusion scheme is implemented on the multi-format and multi-modal combination of tabular borehole data and three land surface raster data sets.
- The FiLM (Feature-wise Linear Modulation) [11] technique was applied to a custom-built deep learning model. The findings revealed the effectiveness of the FiLM technique in scenarios where visual processing conditions the processing of tabular data,

building upon the success previously demonstrated in [11], which explored a distinct scenario where linguistic processing conditions visual processing.

- The study presents the outcomes of ground ice prediction based on the existing data, accompanied by a multi-depth visualization of the predictions. The results include an examination of the absence of correlation between the surface and sub-surface data sets, supported by a statistical comparison to determine the expected correlation.

- Criteria for achieving more accurate ground ice prediction was established, involving the utilization of Shapley analysis to discern which inputs to keep or discard based on their impact. Additionally, methods for summarizing 3-D borehole samples into representative depth profiles are introduced, facilitating future investigations in ground-ice prediction on more macroscopic scales.

Chapter 2

Data fusion and pre-processing

2.1 Borehole Data - Tabular Format

As the aim is to characterize and predict ground ice characteristics in 3-dimensional space, investigations must be based on 3-D borehole data. Although borehole data sets in various regions of Canada are available, many data sets are 2-dimensional in nature, where samples vary in latitude and longitude only. An aggregated collection of the desired 3-D borehole data is available through PermafrostNet ERDDAP [12].

This borehole data can be characterized as "sparse 3D," with sparse sampling in the lateral dimensions (longitude and latitude) and relatively more finely sampled in the depth dimension, as seen in figure 2.1. The spatial distribution in longitude-latitude follows approximately the diagonal shape of the ITH (see figure 2.2), since the boreholes were intentionally drilled in areas of interest.

The borehole data set consists of approximately 2700 samples at various depths taken at approximately 560 borehole cores in the ITH region between 2012 and 2017. Ground ice variables recorded include: frozenness, cryostructure, ASTM2488, material, and amount of visible ice.

Statistical analysis of the spatial distribution of the borehole data, as depicted in figure 2.3, reveals that the majority of boreholes have maximum depths ranging from 5 to 10 meters, and most boreholes are spaced approximately 100 meters apart. This provides insight into the approximate resolution of the 3D data.

From the non-uniformity of the borehole dataset and the limited region it covers it is expected that the set does not cover a wide variety of terrain types. In reference to the Ground Ice Map of Canada (GIMC) [3], all borehole location falls within a single spatial polygon unit; that is, the macro scale of the GIMC cannot give insight into the variability within this study region of much finer scale. It is important therefore that we acknowledge fully these limitations and uncertainties in our analyses and interpretation of the results

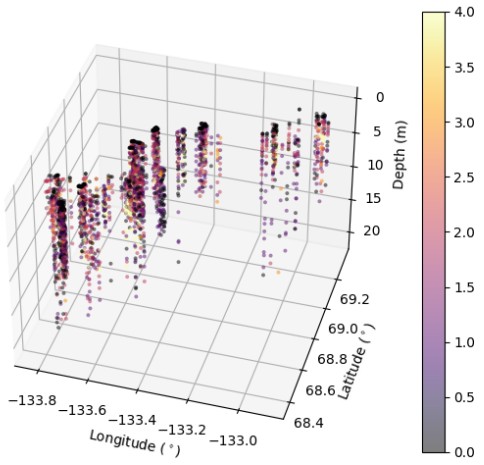


Figure 2.1: Visualization of locations of the 2700-sample borehole data set taken in approximately 560 borehole cores in the ITH region. The colour visualized here indicates the amount of visible ice recorded at each 3D coordinate location. Note that the distribution of the borehole samples is not uniform in any of the three axes.

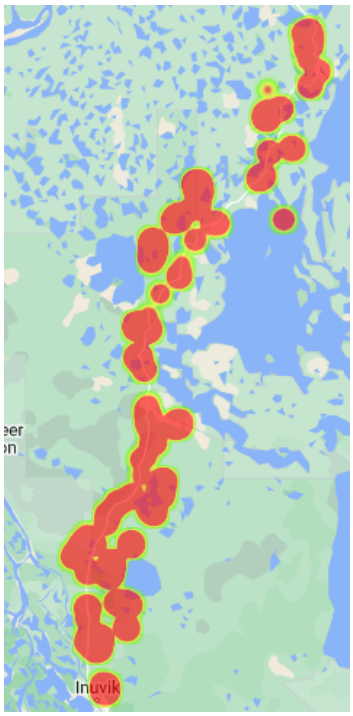


Figure 2.2: Density of borehole samples plotted as a heat map. Borehole locations are observed to be concentrated along the diagonal shape of the ITH.

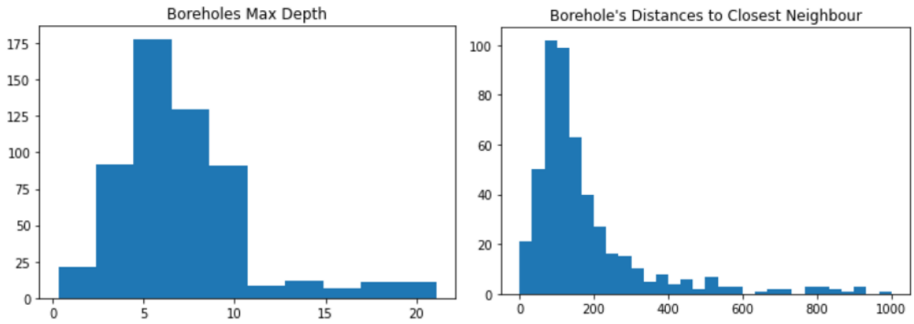


Figure 2.3: Histograms of maximum depth and nearest neighbour distances of boreholes. These figures demonstrate that the majority of boreholes extend as deep as 5 to 10 meters under ground, and the majority of closest neighbour pairs of boreholes are roughly 100 meters apart in longitude-latitude plane.

2.2 Land Surface Data - Raster Format

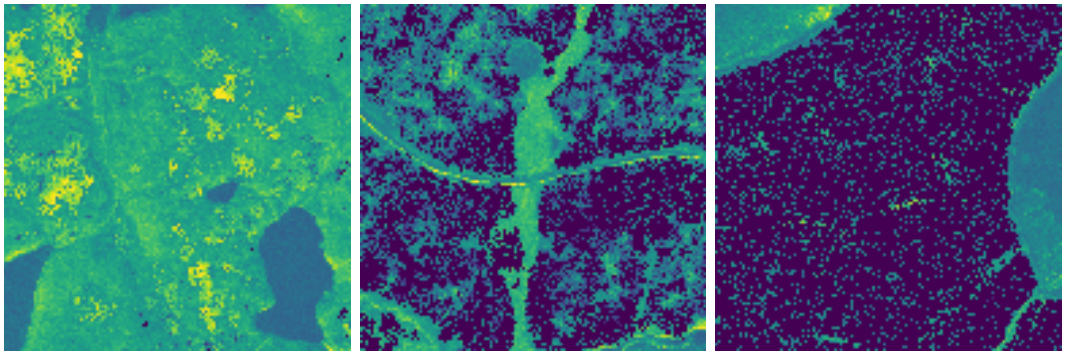
To mitigate the known limitations of the borehole data set, incorporating land surface raster data can provide additional information and help to fill in the gaps in our understanding of the terrain. The raster data can help to identify spatial patterns and relationships that may not be readily apparent from the borehole data alone.

2.2.1 Geomorphometric data at 90m resolution

The Geomorph90m dataset [13] provides a comprehensive set of topographic features at 90-meter resolution for the entire globe. Geomorphometric variables such as slope, aspect, curvature, and various indices of terrain ruggedness and complexity are derived from digital elevation models (DEM). These features can help capture the variation in longitude-latitude plane of ground ice properties across the landscape, as they reflect different aspects of the physical environment that may influence ground ice formation and distribution. Despite its lower spatial resolution compared to the next two data sets, the value of this data lies in the diverse range of the 25 topographic features it offers.

2.2.2 ArcticDEM derived data at 10m resolution

ArcticDEM [14] is a 10-meter resolution digital elevation model (DEM) of the Arctic, produced by fusing and improving satellite stereo images from various sources. Then, SAGA software [15] is used on the DEM to extract and calculate valuable raster variables, including wetness index and solar insulation. Integrating these high-resolution geomorphometric variables into a deep learning model may enhance ground ice characteristic predictions.



(a) High quality chip (b) High quality in the center (c) Low quality chip

Figure 2.4: Neighborhood chips within the HDS-processed SAR raster data exhibit diverse quality levels. Chips with overwhelmingly low quality, as depicted in Figure 2.4c, are systematically discarded. Conversely, high-quality chips, exemplified by Figure 2.4a, are retained. The neighbourhood chips are selected with borehole locations at the center. Notably, when low-quality chips such as 2.4b possess well-defined information at the center, they are retained if deemed useful.

2.2.3 Synthetic Aperture Radar data at 3m Resolution

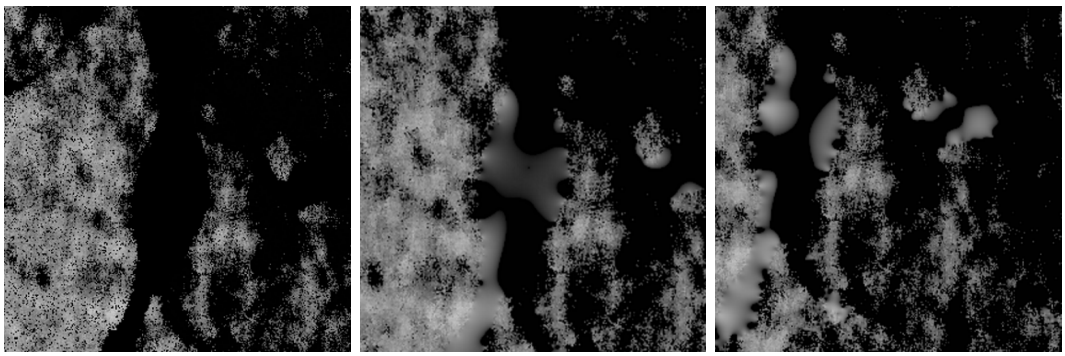
Synthetic Aperture Radar (SAR) is a remote sensing technology that uses radar to capture images of the earth's surface, which can provide valuable information for monitoring changes in terrain conditions. HDS (Homogenous Distributed Scatterer) processing on interferometric SAR (InSAR) image stacks can produce maps of seasonal and long-term surface displacement that provide insight into freeze-thaw cycles of the active soil layer, as well as the degradation or growth of underlying permafrost, particularly in shallow depth layers. Two raster parameters derived from HDS processsing, linear surface motion trend and seasonal surface motion amplitude, will be integrated into the deep learning model.

SAR data pre-processing

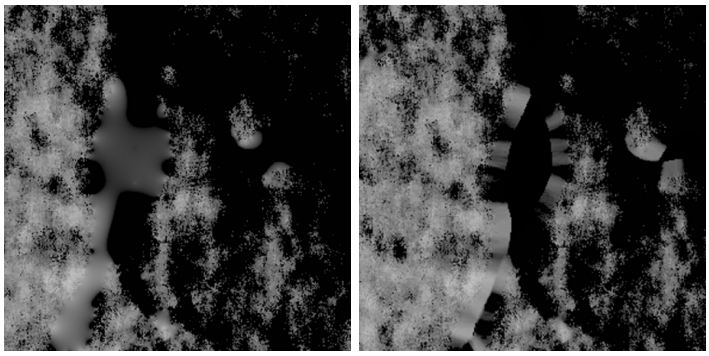
The SAR raster layers contains many null pixels, as seen in figure 2.8a. This non-ideal distribution will affect the unit-variance zero-mean normalization of pixels values, which is needed to pre-process raster layers for input into the deep learning model. Regions containing a majority of null pixels may skew the distribution, hence low quality chips of the raster layers need to be discarded from our experiments, as shown in figure 2.4.

After discarding neighbourhoods of prevalent null values, interpolation algorithms through GAMMA software [16] and GDAL software [17] were employed to fill gaps of null values in the remaining SAR data. Many of these methods created undesirable artifacts through the process of filling gaps, as seen in figure 2.5.

Instead of the gap-filling algorithms presented in figure 2.5, Lanczos is an interpolation method commonly used in image processing to resize or resample images while minimizing aliasing artifacts. The Lanczos interpolation method involves a windowing function known

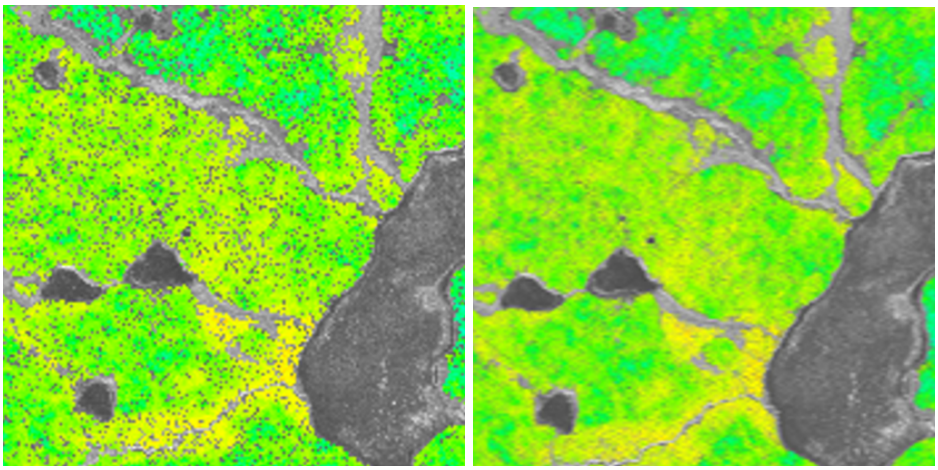


(a) Before (b) Springs interpolation (c) Laplace interpolation



(d) Average of neighbours (e) Adaptive interpolation

Figure 2.5: Various gap-filling algorithms yielded new patterns that could potentially introduce confusion for ResNet. All methods are least-squares algorithms except for average of neighbours.



(a) Before (b) After Lanczos interpolation

Figure 2.6: Before and after lanczos interpolation of SAR raster layer

as the Lanczos window, where the window size denoted by a in the equation:

$$L(x) = \begin{cases} \frac{\sin(\pi x)}{\pi x} \cdot \frac{\sin(\pi x/a)}{\pi x/a}, & \text{if } -a \leq x \leq a, \\ 0, & \text{otherwise.} \end{cases} \tag{2.1}$$

Lanczos interpolation through GDAL is employed to fill null pixels in the SAR raster data, as seen in figure 2.6. Although not perfectly noise-less, when these images are used as input for the ResNet branch in the deep learning model, the convolution processes within ResNet are adept at handling the remaining noise, ensuring robust processing of the input data. This mild gap-filling approach is valuable as it eliminates the high concentration of null values, as illustrated in figure 2.8. Consequently, the zero-mean unit-variance scaling is not negatively influenced by the prevalence null values. In GDAL [17] software, the conversion of coordinate system and resolution, and Lanczos interpolation is done in a single step.

It is worth noting that the available HDS-processed SAR data does not cover the whole region where the borehole data locations reside. Moreover, there are neighbourhoods of the SAR image data that consists of mostly null values, so boreholes in these neighbourhoods must be discarded for experiments involving the SAR image data. When incorporating the HDS-processed SAR data, after discarding low-quality chips as seen in figure 2.4, only around 15% of the the borehole data can be used. This further limitation of data size must be considered when assessing the experiment results involving SAR data in terms of statistical value. Later sections will explore the effect of different data sizes on model predictions.

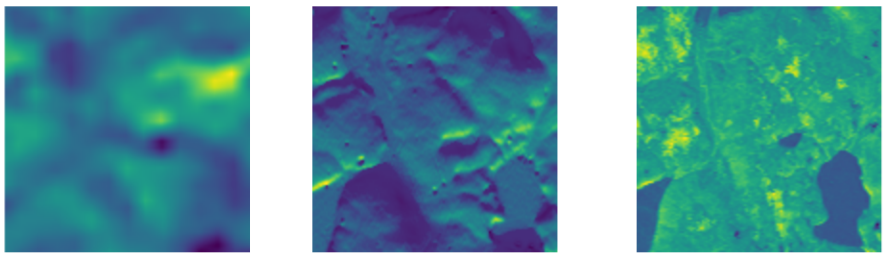


Figure 2.7: Raster layers of the same region surrounding borehole GSC1A - P1

2.3 Data Standardization and pre-processing

2.3.1 Data fusion and normalization

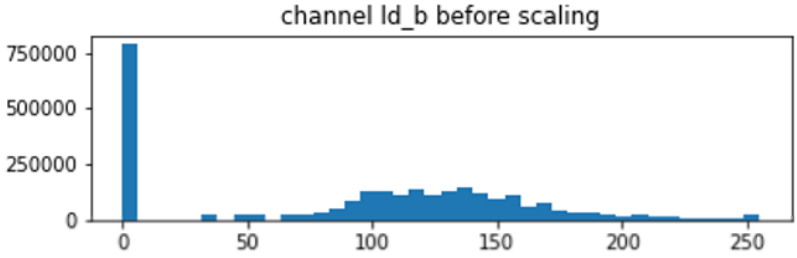
Input raster data

The three raster data sets are interpolated and converted into the same coordinate system, ESPG3413, and the same resolution of 10 meters using GDAL tool. This ensures that they can be interpreted in a standardized manner in the deep learning model.

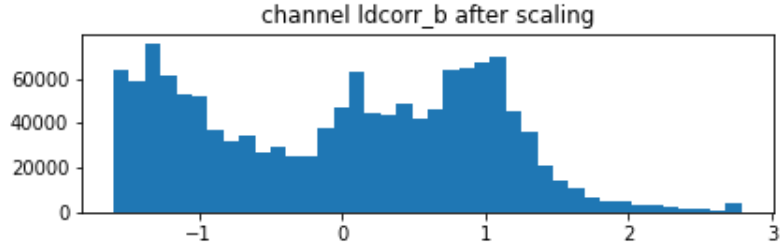
For each of the 30 raster layers, the StandardScaler class from the Python sklearn library is employed to achieve unit variance and zero mean. A key distinction is made between geographic image normalization and RGB image normalization in context of deep learning. Unlike RGB images, where normalization can be done individually for each instance, geographic images require a layer-wise normalization approach. This is because the overall absolute value of an image matters; for instance, a neighborhood with high elevation provides different information than one with low elevation. Therefore, raster layers are normalized to unit variance on a layer basis before being extracted into neighborhood chips. This normalization process also serves to eliminate the magnitude differences between different layers, ensuring a consistent scale across all raster layers in the deep learning model. Figure 2.8 shows the effect of layer-wise scaling on raster data.

Input tabular variables

The longitude and latitude of borehole samples are transformed into ESPG 3413 projection coordinates using the Python pyproj package. This conversion enables the model to extract neighborhood chips from the different sources of raster data in an uniform manner. It's worthy to distinguish between the coordinate system conversion for borehole locations and the raster data. Borehole samples remain fixed in their original locations after conversion, while the raster data undergoes interpolation, resulting in a new grid where the pixels no longer precisely align with their original positions.



(a) Histogram of a SAR raster layer reveals a significant concentration of values at 0, indicating a prevalence of null values.



(b) Histogram of the raster layer after Lanczos interpolation and unit-variance zero-mean scaling.

Figure 2.8: Histograms of a SAR raster layer before and after pre-processing steps. After filling in null values and standardizing the range, the raster layer's histogram no longer displays a prevalent number of null values and is now scaled to unit-variance zero-mean.

Output tabular variables

Subsequently, the tabular variables undergo normalization to unit norm and zero mean before serving as input for the deep learning model. Also, the month parameter has been converted to cyclic numeric values to ensure that January and December are consecutive in value and while winter and summer months are far apart in value.

Categorical variables, like visible ice and materials, are transformed into categorical or ordered categorical codes where applicable. Additionally, new binary variables are generated based on these parameters, such as indicating the presence or absence of visible ice. This facilitates experiments involving binary target values, in addition to the standard multi-category target values.

2.3.2 Training-ready data

Tabular variables that are not ground ice properties are used as input variables for ground ice prediction models:

- longitude
- latitude
- depth

Target variable	Number of classes	Values	Ordered
Frozenness	2	frozen, unfrozen	Yes
Amount of visible ice	5	no ice, low ice,...,pure ice	Yes
Materials	6	gravel, sand, organics,...	No

Table 2.1: Target tabular variables information

Data	Geomorph90m	ArcticDEM	HDS-processed InSAR
Number of Layers	25	3	2
Original Resolution	90 m	10 m	2 m
Original Coordinate System	Longitude-latitude	EPSG3413	SAR-image
Location coverage	100%	100%	15%

Table 2.2: Input raster data information

- length of depth interval
- month
- year

Tabular ground ice properties used as target variables for ground ice prediction models are shown in table 2.1.

The input tabular variables in list 2.3.2, the input raster layers as in table 2.2, and the target tabular varables in table 2.1 combine into a hybrid tabular-image data set, which is adapted to a custom data architecture based on the "Dataset" class of the Pytorch [18] framework.

For each borehole, prediction models (introduced in the next chapter) will extract the corresponding neighbourhood chip containing all layers of the raster data sets. Each extracted neighborhood corresponds to a single borehole location, which houses multiple borehole measurements at various depths. That is, the relationship between the raster data and borehole samples is not one-to-one but rather one-to-many.

Chapter 3

Prediction Models

Now that the data has been packaged and prepared for training, we outline a number of machine learning and deep learning models fit for the task of predicting ground ice.

3.1 XGBoost

XGBoost [19], an efficient and scalable implementation of gradient boosting, stands as a powerful machine learning algorithm widely recognized for its performance and versatility. It excels in handling structured or tabular data, making it a valuable tool for predictive modeling tasks.

XGBoost, or Extreme Gradient Boosting, is an ensemble learning algorithm renowned for its efficiency and accuracy, particularly in tabular data scenarios. At its core, XGBoost builds an ensemble of decision trees sequentially. Each tree corrects the errors of its predecessors, gradually improving the overall predictive performance. The decision trees in XGBoost are shallow, meaning they consist of a limited number of nodes, reducing the risk of overfitting. Its adeptness with tabular variables, handling of missing data, and prevention of overfitting make it a reliable comparator for the other models introduced later in this chapter.

XGBoost is employed as a benchmark for prediction accuracy using only tabular borehole data without the inclusion of raster data. While XGBoost is known for its robust performance on tabular data, it lacks advanced capabilities for the fusion of tabular and image data. As a basic approach to incorporating raster information, we employ a naive fusion strategy by appending the value of the pixel nearest to the borehole location as an additional tabular input variable. The resulting prediction accuracies are presented in table 3.1 for each target variable.

It is observed that the addition of the single pixel does not make the prediction better or worse.

This comparison serves as a reference point for evaluating the effectiveness of our more sophisticated data fusion schemes in the next sections.

Target variable	XGBoost	XGBoost with concatenation	Random chance
Frozenness	0.963	0.963	0.876
Materials	0.683	0.686	0.466
Visible ice	0.503	0.497	0.337
Visible ice (binary)	0.751	0.751	0.663

Table 3.1: Comparison of XGBoost's prediction accuracies between naive data fusion of concatenating the value of the single closest pixel and no data fusion. The addition of the single pixel does not appear to impact the prediction. Random chance accuracy serves as an accuracy baseline of a model with zero predicative power. Model performance is considered meaningful when it surpasses this baseline. See section 3.1.1 for more details on random chance accuracy.

3.1.1 Random chance accuracy

Random chance accuracy serves as a baseline, indicating the accuracy expected from the "best attempt" guess of a blank model which has no predicative power. In general, a model's performance is considered meaningful when it surpasses this baseline, signifying its ability to learn and make predictions beyond random chance.

Random chance accuracy is calculated based on the ratio of sample support for the target classes. For example, consider a target variable with five possible classes: no visible ice, low visible ice, medium visible ice, high visible ice, pure ice. If this data set was an ideal test data set for machine learning, each class would have equal support size, and the accuracy of the random chance guesser would be $1/n$, where n is the number of classes. However, our borehole data set in question does not have even support sizes, The random chance guesser would have the best accuracy if it guessed all the samples were of the class with the largest support size. Therefore, the random chance accuracy would be

$$\text{random chance accuracy} = \frac{\text{support of the largest class}}{\text{total size}} \tag{3.1}$$

3.2 Application of FiLM: a conditioning normalization technique for deep learning models

Various machine learning architectures are available to train on the borehole data alone, such as decision trees and the classic feed-forward neural networks known as multi-layer perceptron (MLP) [7]. However, these methods cannot naively take land surface raster data as input in addition to the borehole variables.

The objective here is to augment the knowledge of a deep learning model such as an MLP by incorporating the raster image data, in addition to the borehole tabular data. One

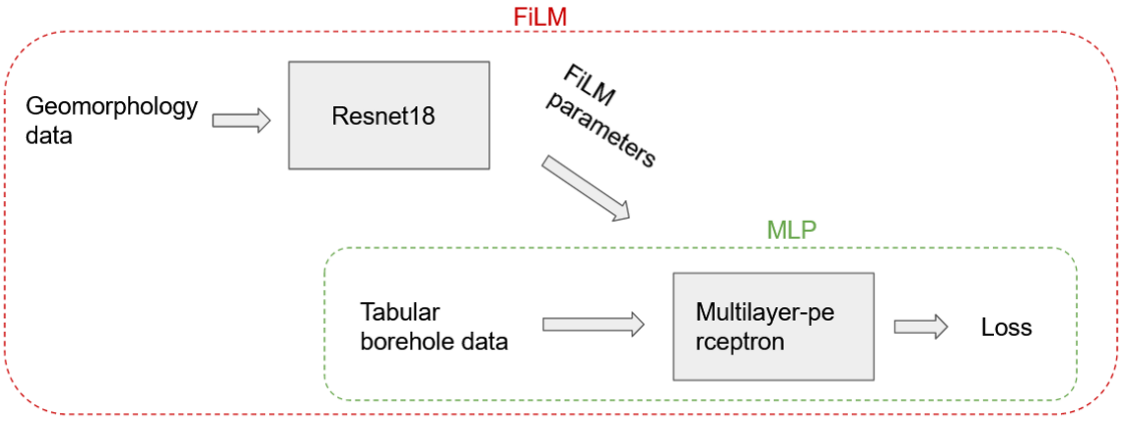


Figure 3.1: FiLMed (feature-wise linear modulation) deep learning architecture, where the borehole data is ingested through the main branch and the geophysical raster data is ingested through the side branch.

approach to combining multi-modal data involves using a custom deep learning architecture where the borehole data is ingested through the main branch and the geophysical map raster data is ingested through the side branch. The purpose of the side branch, also known as the generator branch, is to output a set of scaling and shifting parameters that condition the network on the additional raster information.

The adoption of the FiLM (Feature-wise Linear Modulation)[11] technique in our model is motivated by its capacity to generalize effectively from limited data to more intricate scenarios not encountered during training. This ability is crucial for adapting to diverse and complex real-world conditions. The FiLM approach provides the model's generator with moderately fine-grained control over activations at each FiLM layer, enabling nuanced adjustments in response to varying inputs. Additionally, the conditioning mechanism of FiLM is computationally efficient, with a computational cost that does not scale with image resolution. This efficiency is particularly advantageous, as it allows for effective modeling of high-resolution imagery without imposing significant computational burdens.

3.2.1 FiLMed MLP architecture

As shown in figure 3.1, the main branch has a MLP structure and ingests the tabular borehole data, and the side branch has a convolution neural network (CNN) structure known as Resnet [6], which is commonly used to solve computer vision tasks.

This custom architecture is implemented based on Feature-wise linear modulation (FiLM) [11] technique, where FiLM layers influence model computations in simple affine transformations applied on the network's intermediate layers. An affine transformation can be interpreted as a geometric transformation that preserves lines and parallelism. The FiLM layer transformation is defined as follows[11]: for the ith input's c^{th} feature, given a neural network's activations $F_{i,c}$ (which are intermediate layers in the main branch),

$$FiLM(F_{i,c}) = \gamma_{i,c}F_{i,c} + \beta_{i,c}, \tag{3.2}$$

where $\gamma_{i,c}$ and $\beta_{i,c}$ are parameters generated from the side branch of the network architecture. It is worth noting that although the FiLM parameters $\gamma_{i,c}$ and $\beta_{i,c}$ are called parameters, they are not the same as the typical function parameters learned by a deep learning network during training. The FiLM parameters are considered as outputs to the side branch, and as function parameters to the main branch. The main and side branches are trained simultaneously through a single loss function.

The number of parameters generated by the Resnet side branch depends on the number of FiLM normalization layers inside the MLP, and the number of blocks and width of the MLP. The model incorporates early stopping based on validation accuracy, with a maximum training limit of 50 epochs. The training and testing program utilizes 5-fold cross-validation. The prediction accuracy and standard deviation are estimated by averaging over multiple experiments.

3.3 Verification of FiLM Technique

In conventional visual question answering pipelines, integration of language and image information is achieved through methods like element-wise product, concatenation, or multi-step reasoning. The FiLM technique differs in utilizing one source of input data as modulation parameters rather than features. Notably, the original FiLM technique has been tested on the setup where the linguistic-processing side branch conditions the visual processing main branch. Meanwhile, the custom architecture employed in this project involves the visual-processing side branch conditioning the main branch that handles tabular data. Given these deviations from typical multi-modal data fusion and the previously validated FiLM setup, it becomes imperative to verify the effectiveness of our custom FiLM architecture.

The validation of this customized FiLM model is conducted using a modified version of the CIFAR-10 test dataset.

3.3.1 Synthetic tabular variables

The CIFAR-10 dataset in its original form comprises RGB images as input data, each associated with labels representing 10 classes, such as airplane, bird, cat, truck, etc. Given that our FilM'd model requires both image and tabular data as input, a process was initiated to generate the necessary tabular data used as input for the main branch of the model.

A good candidate of the synthetic tabular data would be meta data giving background information on the class. The chosen variables are "cargo capacity", "number of legs", "number of wheels", "is organic", and "Water Mobility". Table 3.2 describes synthetic tabular input variables for the 10 classes of CIFAR10.

Class	Cargo capacity	Num of legs	Num of wheels	Is organic	Water mobility
Airplane	1000	0	10	0	0
Automobile	200	0	4	0	0
Bird	0	2	0	1	0
Cat	0	4	0	1	0
Deer	0	4	0	1	0
Dog	0	4	0	1	1
Frog	0	4	0	1	1
Horse	100	4	0	1	0
Ship	5000	0	0	0	1
Truck	500	0	4	0	0

Table 3.2: Values of synthetic tabular data of CIFAR10 classes

Model	Num of training samples	Prediction Accuracy
Non-FiLM	10000	0.80
FiLM	10000	0.88
Non-FiLM	5000	0.70
FiLM	5000	0.86

Table 3.3: Prediction accuracy of FiLM verification experiments

Note that certain classes, such as Cat and Deer, share identical synthetic variables. Consequently, a model relying solely on the synthetic tabular parameters may struggle to distinguish between these classes, resulting in imperfect predictions, which leaves room for potential improvement by incorporating image data.

The FILM verification experiment aims to assess whether the incorporation of the side branch, which processes image data containing valuable information like the differentiation between Deer and Cat, enhances prediction accuracy compared to a model without a side branch.

3.3.2 Verification successful

In comparing the prediction accuracies of the FiLMed and non-FiLMed models across different training sample sizes, it was observed that the FiLMed model consistently outperforms the non-FiLMed model, as seen in table 3.3. These results suggest that the FiLMed network effectively captures useful information in additional data processed through the side branch. The model's ability to enhance accuracy, even with a smaller training dataset, displays its potential to identify meaningful correlations in land surface raster data (if meaningful correlations is present in the data), ultimately improving predictions of ground ice characteristics.

3.4 Reason for using FiLM

In this project, FiLM technique is utilized to incorporate two modes of data - tabular borehole data and geophysical raster data - into a single neural network. The tabular data is ingested through the main branch of the network while the image data is ingested through the side branch. The FiLM layer is used to condition the network based on land-surface information, which may influence ground ice parameter predictions. By combining these two modes of data, the network can learn to better capture the complex relationships between the sub-surface ground ice properties and the surrounding surface characteristics.

Chapter 4

Results and analysis

4.1 Results

Table 4.1 provides a comparison of prediction accuracies and their standard deviations for different models introduced in the previous chapter, specifically focusing on predicting the visible ice target variable.

The predominant class in the visible ice variable is "no visible ice," constituting around 33% of all samples. A model with no predictive power, such as one classifying all samples into the "no visible ice" category, would have a prediction accuracy of 0.33—representing the random chance accuracy benchmark for visible ice 3.1.1.

While all models exhibit some predictive power (accuracy greater than 0.33), the general trend suggests that the addition of raster data does not enhance prediction accuracy.

4.2 Potential factors in lack of correlation found

Contrary to the verification experiments, incorporating raster data through the FiLM architecture appears to lower the prediction accuracy. There may be a number of factors at play here.

Model	Raster Data used	Location Coverage	Accuracy	Std Dev
XGBoost	None	100%	0.50	0.022
MLP	None	100%	0.49	0.020
FiLM	Two geomorph datasets	100%	0.42	0.014
MLP	None	15%	0.48	0.037
FiLM	All three datasets	15%	0.40	0.069

Table 4.1: Comparison of prediction accuracy of various methods on validation sets.

Model	Chip size	Prediction accuracy
FiLM	32	0.42
FiLM	64	0.44
FiLM	128	0.49
Non-FilM	N/A	0.49

Table 4.2: Comparison of prediction accuracy of various neighbourhood chip size of raster layers.

4.2.1 Effect of neighbourhood chip size

A borehole contains samples at multiple depths. For each sample, the FiLMed deep learning model (figure 3.1) extracts a neighbourhood of a desired chip size surrounding its longitude-latitude location from the collection of raster layers 2.2. An investigation is conducted to determine if the chip size of this neighborhood affects the model's predictions. The raster layers, standardized to a 10-meter resolution, are experimented with using square chip sizes ranging from 32 x 32 pixels to 128 x 128 pixels. Table 4.2 presents the findings.

The analysis reveals that prediction accuracy tends to improve with larger chip sizes, yet it doesn't surpass the non-FiLMed model. The effectiveness of the raster layers in enhancing prediction remains uncertain.

4.2.2 Insight from ICA

Independent Component Analysis (ICA) [20] is a mathematical technique designed to uncover statistically independent components within a set of observed multivariate data. The fundamental concept behind ICA involves the decomposition of a mixed signal into a linear combination of non-Gaussian source signals. Mathematically, this can be expressed as:

$$x = As, \tag{4.1}$$

where x represents the observed mixed signal, s is the vector of the true independent source signals, and A is the mixing matrix. The goal of ICA is to estimate the inverse of the mixing matrix, denoted as W, where then an approximation of the independent sources $\hat{s}$ can be calculated:

$$\hat{s} = Wx \tag{4.2}$$

The estimation of W involves maximizing the non-Gaussianity or independence of the estimated source signals.

Certainly, upon completing ICA, the obtained approximated independent sources $\hat{s}$ and the approximated mixing matrix $\hat{A}$, where $\hat{A}$ is the inverse of W, can be leveraged to reconstruct an approximation of the original observed signals:

$$\hat{x} = \hat{A}\hat{s} \tag{4.3}$$

Compared to the well-known Principal Component Analysis, ICA is more suited to tasks such as blind-source separation. In context of the borehole data, the ICA method is applied to compute dominant borehole profile prototypes. The vertical profile of a borehole in the ITH data can be interpreted as a mixed signal - a linear combination of a number of dominant vertical profile prototypes. This analysis informs the longitude-latitude spatial distribution of boreholes, using ICA to reduce the 3-dimensional borehole data to 2 dimensions, compressing the depth axis. This 2D spatial distribution can be then correlated and analyzed with the 2D land surface raster data discussed in subsequent sections.

Before applying ICA, the borehole data set is interpolated into a uniform 3D grid, spanning 0-5 meters in the depth dimension at 0.5 meter grids. The FastICA [21] class from the Python sklearn.decomposition library is employed to uncover the underlying independent sources in the interpolated 3-D borehole data.

In employing FastICA, the user has the responsibility of specifying the number of independent sources, an important decision to achieve a reasonable balance between approximation accuracy and convergence. As the number of components increase, Figure 4.1 reveals a decrease in the mean-square error of ICA approximation. However, the fastICA algorithm does not converge for $n = 5$ and $n = 6$ after 200 iterations, suggesting that fewer components might be more effective in balancing accuracy, convergence, and potential overfitting.

Figure 4.2 visualizes the independent borehole profile sources alongside their approximations of the original observed signals for the number of components $n = 2$ and $n = 3$. In this context, selecting $n = 3$ seems to strike a suitable compromise.

2-D spatial mismatch

The purpose of deconstructing borehole data into sources and mixing components through ICA is to enable the summarization of each borehole location, housing multiple samples at various depths, into a single value. Each borehole is then classified into the category of the borehole profile it is most dominated by.

The ICA algorithm compresses the 3-D data into 2D, and clustering is performed on raster data chips to categorize them into classes using Resnet architecture, a method that had previous success in remote sensing applications [8, 9].

The results in figure 4.3 reveal a mismatch in statistical properties between the borehole and raster data, possibly explaining why the addition of extra raster data to the FilMed

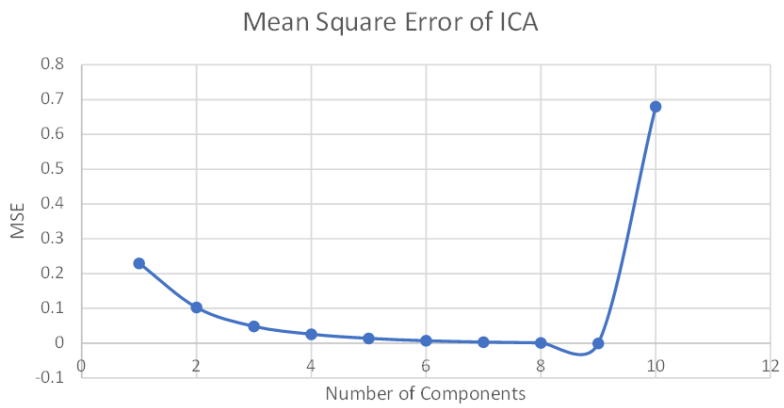
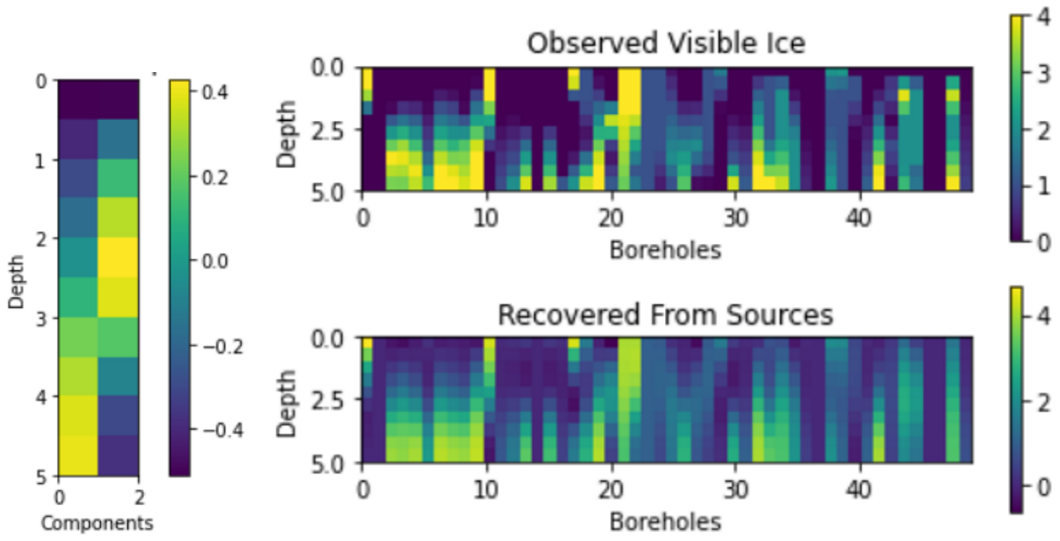


Figure 4.1: Mean-square error of ICA approximation as expressed by equation 4.3. The trend indicates improved accuracy with a higher number of components, but a higher risk of over-fitting.

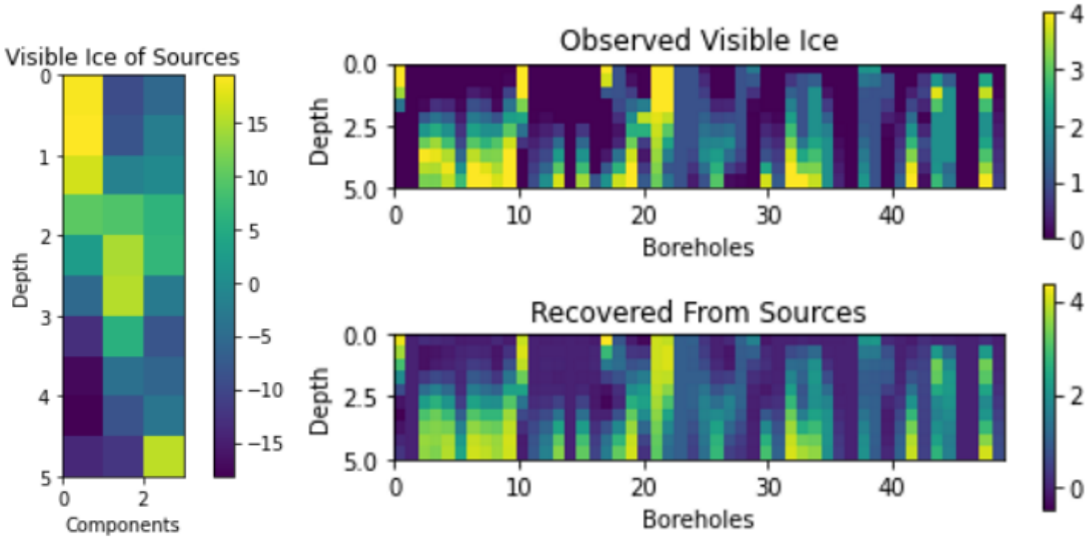
deep learning model did not enhance prediction accuracy. The observed mismatch may be attributed to the inherent differences in the scale and structure of the datasets. While the borehole data exhibits more pronounced variations in depth rather than spatially in terms of longitude-latitude 2-D plane, raster data primarily captures the 2-D spatial variability. The challenge arises from the fact that the high variability of borehole data at fine scales in depth could potentially mask any existing correlations with the 2-D spatial patterns explained by the raster data. The next chapter will explore potential ways to navigate this challenge of scale and variability mismatch.

4.3 Interactive multi-depth visualization of ground ice properties

The Google Earth Engine Webapp is a convenient visualization tool that displays the ground ice predictions on the ITH region. It allows users to easily toggle between viewing multiple depth layers, as seen in figure 4.4, and viewing layers at various opacity to compare with the surrounding circumstances of the map, providing an intuitive way to explore and analyze the spatial distribution of ground ice properties. For example, figure 4.5 shows that the prediction made by the MLP model matches physical observations in that the lower left corner of the grid of prediction consists of forestry, therefore there is the least amount of visible ice in that region, as visualized by the black colour.



(a) Sources $n = 2$ (b) Approximation $n = 2$

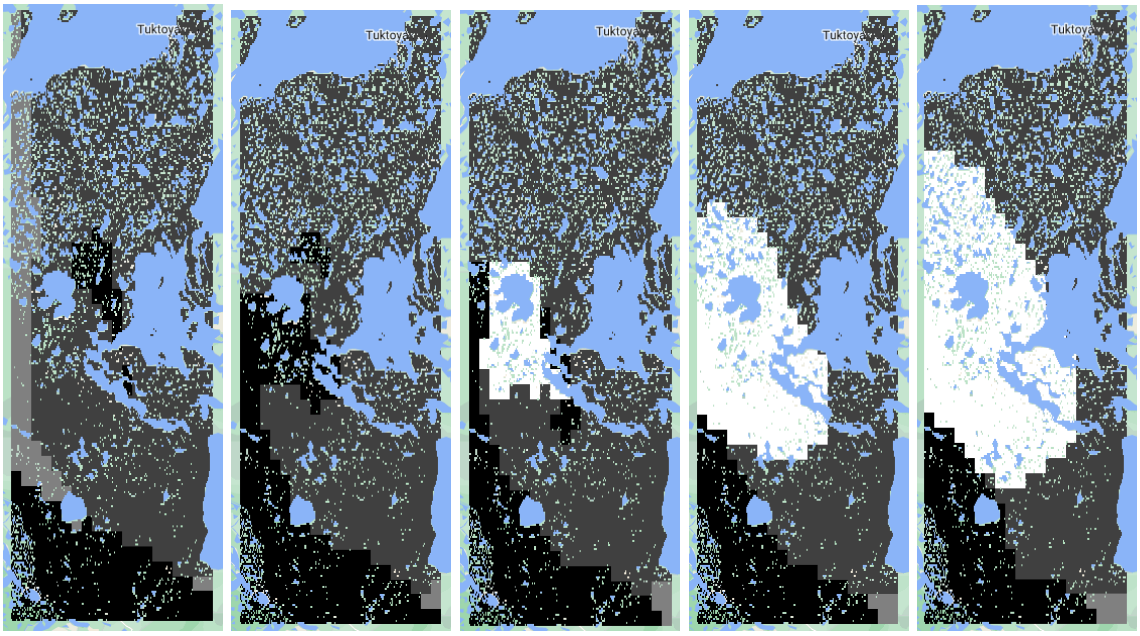


(c) Sources $n = 3$ (d) Approximation $n = 3$

Figure 4.2: ICA-computed sources of borehole profile for the visible ice parameter, where each column in the sources is a vertical borehole profile pattern. The colour shows the trend of the amount of visible ice in the borehole profiles. The $n = 3$ approximation is more accurate than the $n = 2$ approximation as shown in higher similarity between the observed and recovered visible ice profiles.



Figure 4.3: Comparison of the 2-D (longitude-latitude) variability between borehole data (left) and raster data (right) highlights a noticeable mismatch, providing a plausible explanation for the FiLM model's inability to establish a correlation between surface and subsurface data.



(a) 0-2m (b) 2-4m (c) 4-6m (d) 6-8m (e) 8-10m

Figure 4.4: Visualization of visible-ice predicted by MLP model for each 2 meter thick depth layer. The greyscale colour indidates how much visible ice is predicted, with black being no visible ice, and white being pure ice.

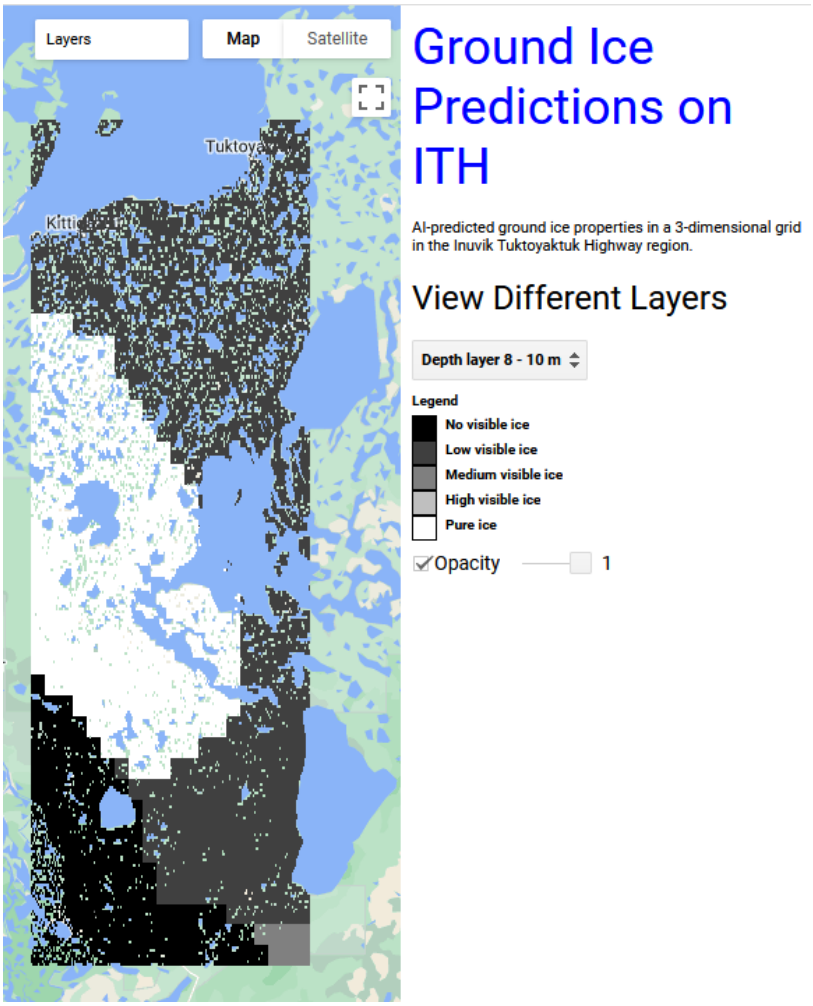


Figure 4.5: Webapp on Google Earth Engine display ground ice properties at various depths. The particular depth interval displayed can be changed through drop down selection. The greystale colour cooresponds to how much visible ice there is expected to be at each layer and location. URL: `https://bqzhang.users.earthengine.app/view/ground-ice`

Chapter 5

Recommendations for future investigations

Understanding the credibility of predictions from complex deep learnign models can be challenging. Various analysis methods are employed to explore the underlying probability distribution of data variables and assess the predictive capabilities of model inputs. In the context of mapping ground ice, the following recommendations emerge from the analyses:

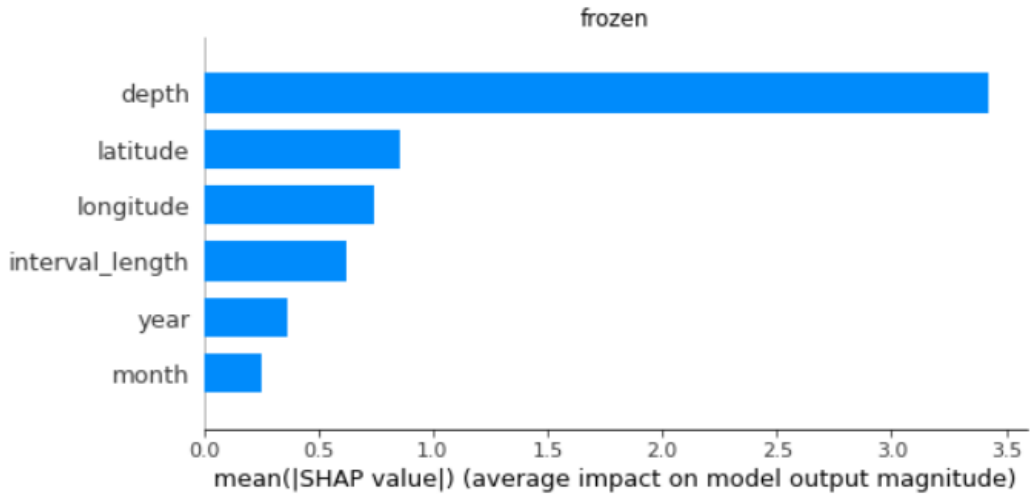
- Feature importance analysis may guide future borehole investigations by identifying the most influential surface characteristics to sub-surface properties.
- Summarizing 3D borehole data into 2D depth profiles on macro scales may alleviate the challenge of scale and variability mistmach

5.1 Feature importance evaluation through SHAP

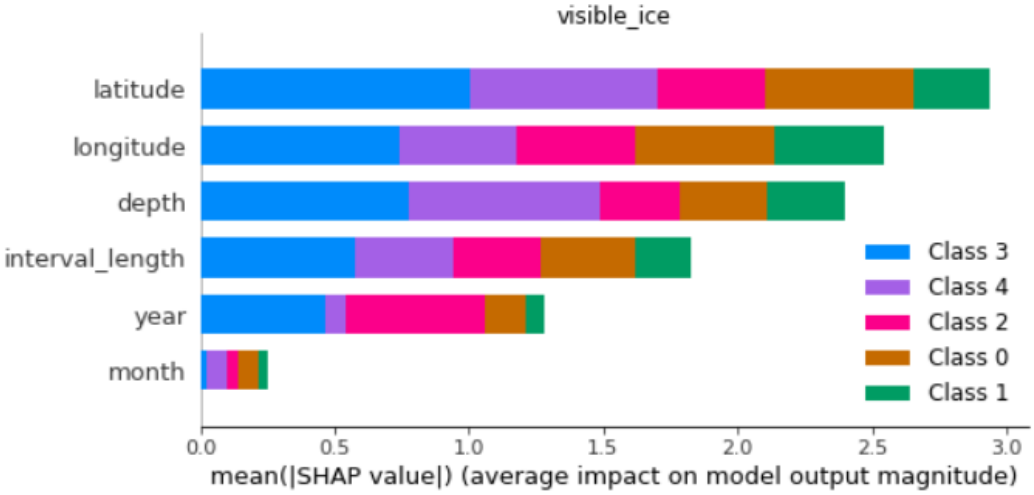
SHapley Additive exPlanations(SHAP) [22, 23] is a framework that interprets predictions by assigning importance values to features to identify the most crucial predictors in a given data set, and it is not limited to tabular data but can also analyze raster data.

The application of the SHAP tool to experiments involving only tabular borehole data through XGBoost predictor reveals that depth is the most influential variable, as seen in figure 5.1. This is likely due to greater variability of ground ice properties with respect to depth compared to with respect to other variables.

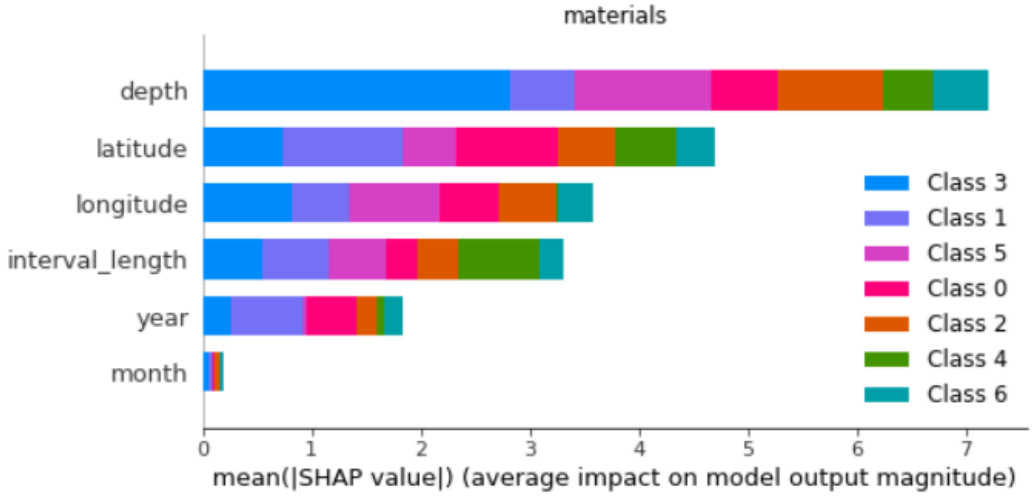
Shap.DeepExplainer is an advanced algorithm designed to estimate SHAP values for deep learning models, building upon the principles of the DeepLIFT algorithm by incorporating per-node attribution rules to approximate Shapley values. Unfortunately, it cannot be applied directly to our FiLMed model or its separate branches. DeepExplainer explains how each input contributes to each of the outputs, making it challenging to interpret in cases where the inputs are intertwined through the parameter-generation and linear modulation processes. However, it could be effective for a simpler deep-learning classifier like



(a) Frozenness



(b) Visible Ice



(c) Materials

Figure 5.1: Feature importance analysis of the tabular input variables with SHAP tool. The coloured bars indicate how much each feature contribute to each target class.

ResNet, where DeepExplainer can elucidate the contribution of each input raster layer to each of these classes.

For this ResNet experiment, we utilize the primary borehole profile identified in the previous ICA investigations with $n = 3$ as categorical labels, as seen in figure 4.2c. That is, the target label of each borehole is the categorical class of the depth profile it is most dominated by.

The raster layers serve as the input data for the model, with the aim of learning the relationship between the 2-dimensional input data and the 2-dimensional target data. It's important to note that this setup differs from the FiLM configuration, which involves 3-dimensional data. Among the ResNet architectures, ResNet18 is selected due to the relatively small size of our dataset, comprising 560 boreholes profiles whose vertical variability has been dimension-compressed by ICA.

The raw SHAP values from the DeepExplainer output, displayed in figure 5.2, reveal more positive correlation in red pixels and more negative correlation in blue. Larger magnitudes of SHAP values indicate more influential pixels. To quantify the overall impact, we sum the pixel SHAP value magnitudes for each chip and average across the validation dataset (approximately 100 boreholes). Figure 5.3 illustrates the predictive power of each raster layer for the three ICA-derived representative borehole profiles.

The SHAP tool provides quantitative measures of the importance of each input feature, which helps to identify which layer are most valuable and optimize future investigations. In consideration of future borehole measurements, it would be beneficial to document on-site surface properties resembling the top raster variables. Despite the availability of Digital Elevation Model (DEM) and remote sensing data, cross-referencing with on-site recordings could enhance the comprehensiveness of a dataset. The top five influential raster variables are defined below according to the data sets [13, 14]:

- **dev-scale**: deviation scale. Deviation refers to deviation from mean elevation and scales refers to the window size used in the multi-scale estimation of spatial patterns.
- **aspect**: the angular direction that a slope faces
- **tcurv**: tangential curvature. It is the rate of change of elevation perpendicular to the slope gradient, and it is related to the convergence and divergence of flow across a surface.
- **twi**: topographic wetness index, the tendecy of an area to accumulate water.
- **tpi**: topographic position index, the difference between the elevation of a focal cell and the mean of its 8 surrounding cells.

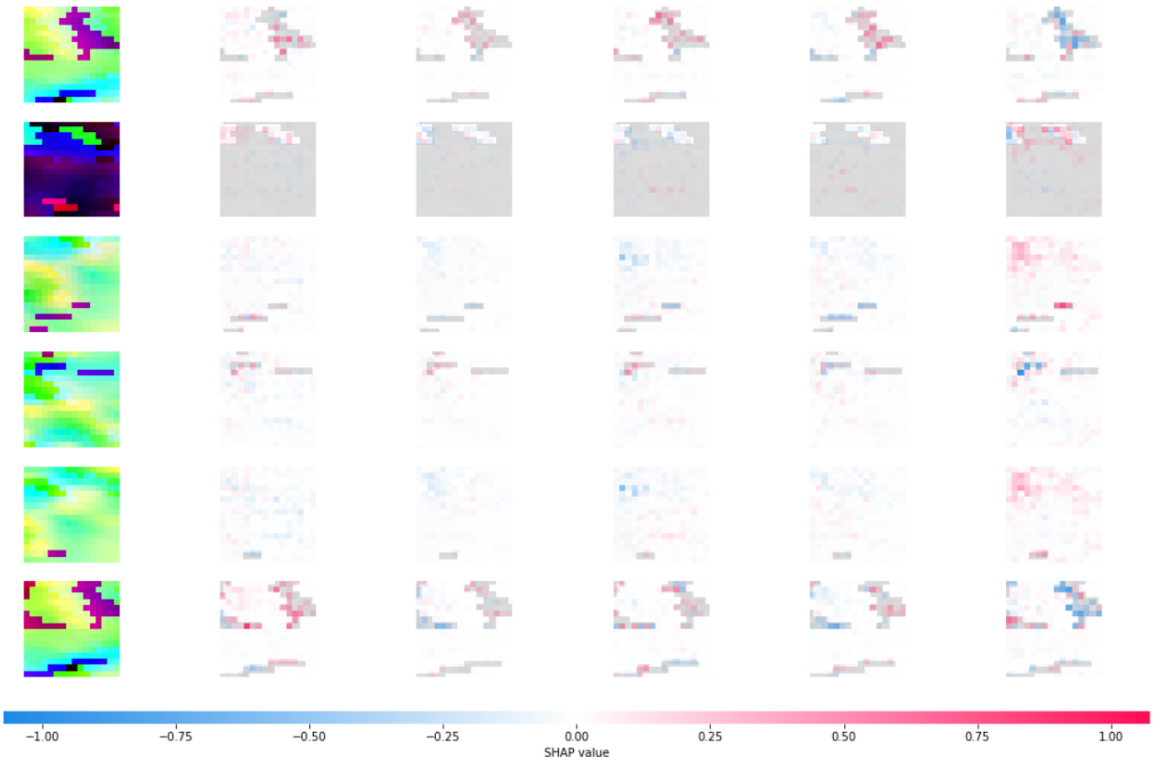


Figure 5.2: Raw SHAP values for each pixel of each raster chip. The first column contains the image chips extracted by the model, and the following columns show their relationship with the target classes. The red colour in pixels signifies positive correlation and the blue colour signifies negative correlation. The magnitude of these pixels values are averaged over all pixels in each raster layer to produced the next figure.

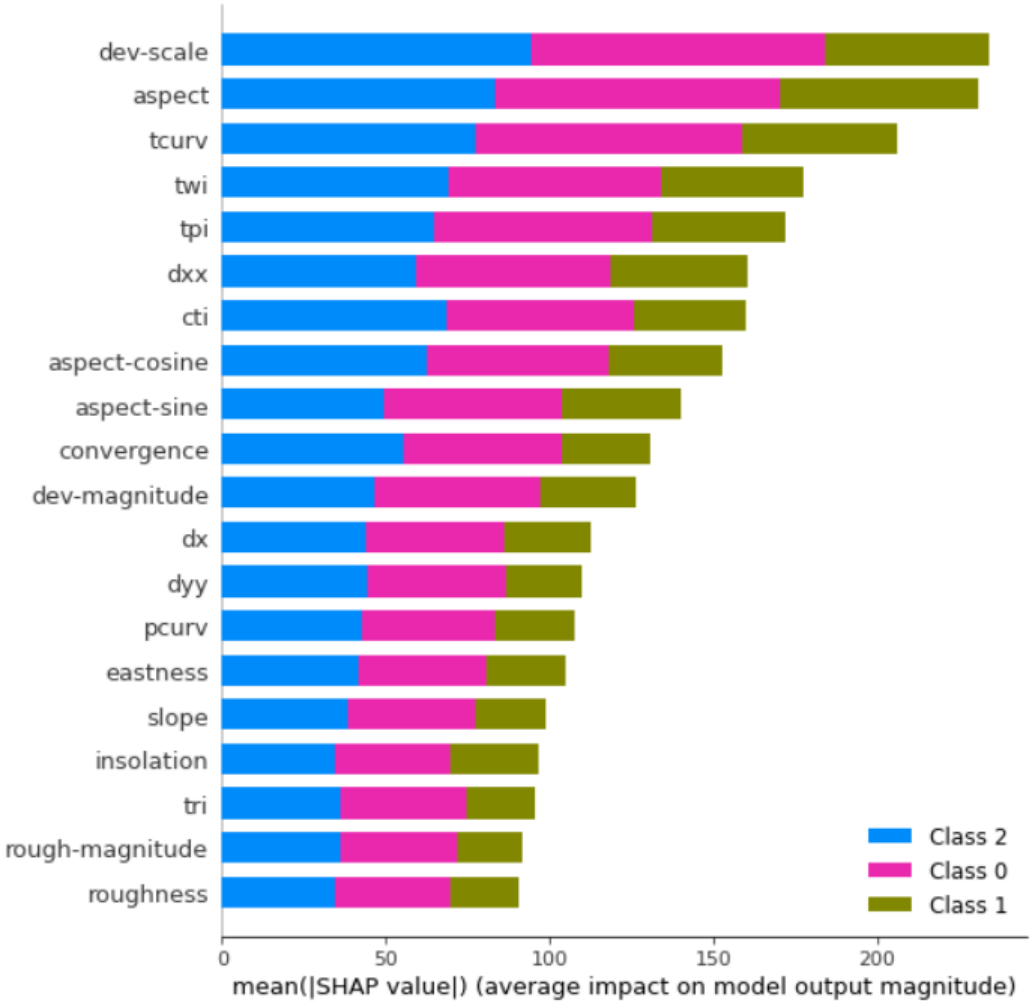


Figure 5.3: SHAP values for input raster layers, indicating their impact to the model prediction. The coloured bars indicate how much each raster layer contributes to accurate prediction of each target class.

5.2 Summarizing borehole profiles on a more macro scale

On a more macroscopic scale, we aim to find a solution capable of summarizing multiple multi-depth borehole measurements into coherent representative distributions of borehole depth profiles.

In the context of sparse 3D borehole measurements within an irregularly shaped area, the objective is to partition the 3-D borehole data into distinct layers in depth, with each layer exhibiting similarity in properties. Although the previously discussed ICA is capable summarizing borehole profiles, it is more suited to summarize continuous properties such as the amount of visible ice, and not unordered categorical properties such materials.

Deciding what depth to divide at depends on multiple variables, such as visible ice, and materials. This process resembles a decision boundary problem, wherein each variable acts as an axis, and the decision boundary manifests as a straight hyperplane intersecting the depth axis.

Calculating the decision boundary involves logistic regression. Higher-order logistic regression introduces decision boundaries that can curve and are not strictly linear. To have a neat partition of 3-D borehole data in depth, the decision boundary needs to take the form of a hyperplane that is as orthogonal to the depth axis as possible. A hyperplane, in this context, is defined as a subspace with a dimension one less than that of its ambient space. For example, a hyperplane in 2-D space would be a straight line, and a hyperplane in 3-D space would be a plane.

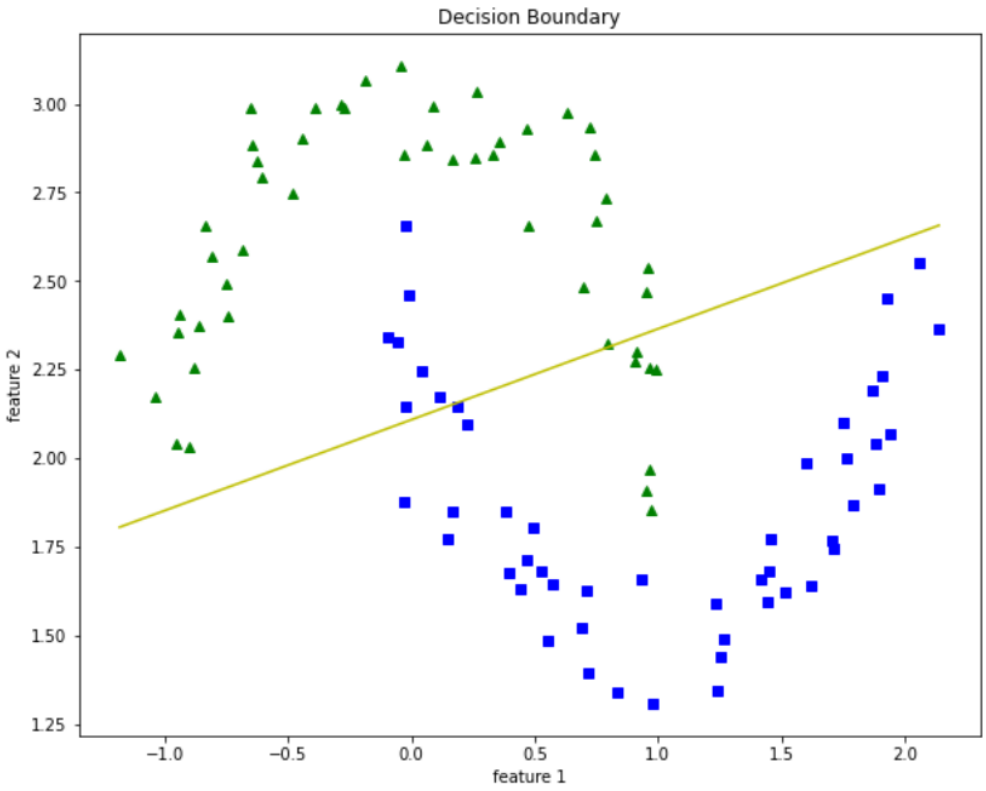
5.2.1 Modified logistic regression on simplified data

The feasibility of logistic regression is assessed in generating a linear decision boundary that adequately addresses our requirement for a partition orthogonal to the depth axis. To simplify the analysis, the preliminary experiments focus on two-dimensional data with one-dimensional label, wherein the target label represents one of the ground-ice properties, and Feature 1 encapsulates all other characteristics influencing the similarity or dissimilarity of data points, and Feature 2 represents depth. The fundamental challenge now becomes whether the decision boundary produced by logistic regression can maintain a reasonable level of orthogonality to the depth axis.

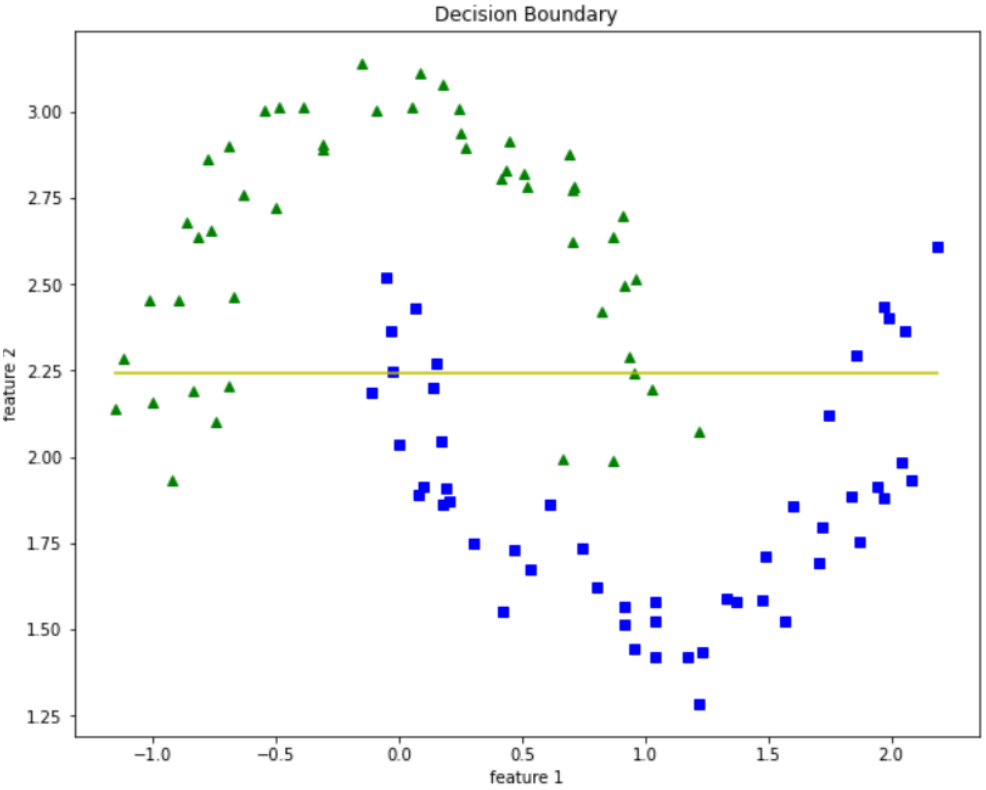
To test this, a 2-D test data set was generated using the "make moons" function from the sklearn.datasets library in Python. The "make moons" function creates two data sets with different labels in the form of interleaving half circles, as seen in figure 5.4.

An accurate decision boundary solution to this dataset would not be linear, it would need a curve to that goes in between the two crescent-shaped classes. This dataset is useful in terms of assessing how logistic regression methods balance between accuracy and how horizontal the decision boundary is.

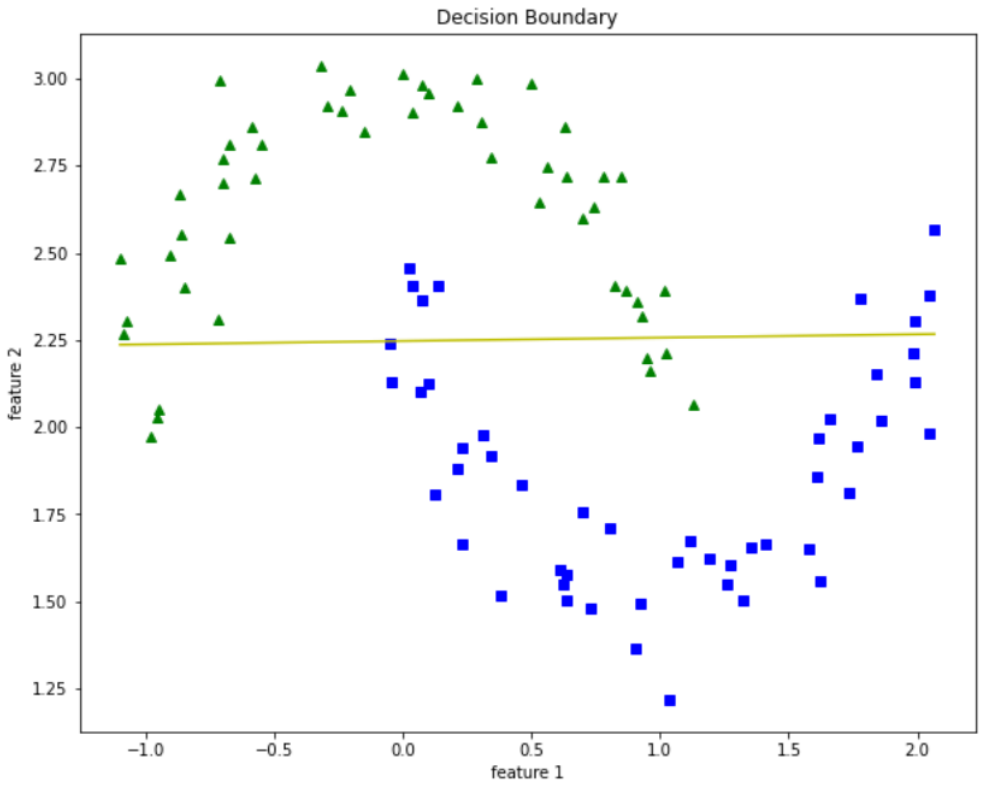
Figure 5.4 shows various modification versions on logistic regression on this test data.



(a) Basic logistic regression. Accuracy = 0.86



(b) Manually set slope to 0. Accuracy = 0.79



(c) Regularization from slope magnitude. Accuracy=0.85

Figure 5.4: Comparison of various modifications to logistic regression with the objective of maintaining the decision boundary as horizontal as possible. The third method adds a regularization term to the gradient based on the the slope magnitude, which punishes the training model more severely the steeper the slope is. This approach strikes a reasonable balance in the trade-off between accuracy and how horizontal the decision boundary is.

Regular logistic regression

The first version is regular logistic regression. Since the goal is to have the decision boundary be as orthogonal as possible to the depth axis, we are interested in processes during training that affect the slope of the decision boundary. The only value that changes the slope is the gradient of loss function with respect to the weights.

For m defined as the number of training samples, w the weights, b the bias, X the 2-D input (with depth as the second feature), and y the target, the gradient of loss (L) with respect to weights is

$$\frac{\partial L}{\partial w} = \frac{(\hat{y} - y)X}{m}, \text{ where} \tag{5.1}$$

$$\hat{y} = \sigma(wX + b), \tag{5.2}$$

$$\sigma(x) = \frac{1}{1 + e^{-x}}. \tag{5.3}$$

This gradient is used to obtain the next iteration of weights through

$$w = w - lr\frac{\partial L}{\partial w}, \tag{5.4}$$

where lr is the learning rate

As seen in sub-figure 5.4a, regular logistic regression has tilted the straight decision boundary to achieve higher accuracy.

Manually set slope to 0

The second version is a manual restriction of the slope to zero at each training step. For 2-D data in this case, slope of the decision boundary is directly related to the first dimension of the weights, that is, the slope is zero if and only if the $w[0]$ is zero. Note that the weights and bias are initialized to zero. We manually set $\frac{\partial L}{\partial w}[0]$ to zero every training step, to ensure $w[0]$ stays zero.

As seen in sub-figure 5.4b, the manual restriction on the decision boundary slope only allows the decision boundary to translate vertically in the training process.

Regularization with slope magnitude

Third version is adding a regularization term to the loss function that punishes the loss based on the magnitude of the slope. We add the term $\frac{1}{2}w[0]^2$ to the loss function. This results in the added term $w[0]$ in the gradient of loss with respect to weights. Therefore, the step to get the next iteration of weights becomes:

$$w = w - lr\left(\frac{(\hat{y} - y)X}{m} + w[0]\right), \tag{5.5}$$

As seen in sub-figure 5.4c, the regularization keeps the decision boundary fairly horizontal, and has some capacity to balance between accuracy and the condition of orthogonality to depth axis. However, the setup of having one of the ground-ice properties as the label, and others properties in the feature dimension, is difficult to generalize. In particular, it's difficult to justify which ground-ice property among the candidates (visible ice, frozenness, materials) should be the label.

5.2.2 Borehole data in Lac de Gras region

Another approach is to compress all ground ice properties (visual ice, materials, etc) into one variable, and perform 1-D logistic regression, where depth is the only independent variable, and the label encompasses all dimensions of ground-ice properties. This approach eliminates concerns about which specific ground ice property is used as the label.

There are a number of ways to dimension-reduce tabular data. One way is through principle component analysis (PCA), which reduces data dimensionality to the combined axis of greatest variance. Another way is mean-shift clustering, which conveniently produces the categorical 1-D labels needed for logistic regression.

We work from 3-D borehole data in the Lac De Gras region provided by [1]. The dataset under consideration consists of 24 boreholes, each encompassing measurements at multiple depths, totalling to approximately 200 measurements. An important aspect of this dataset is that each borehole location is associated with a recorded terrain type, classified into one of four categories: till, valley, organics, and esker. This classification enables separate studies for each class and allows the creation of four vertical borehole stratification profiles corresponding to the four terrain types.

The stratification of layers in depth is achieved through a two-step process. Initially, for each terrain type, a clustering algorithm is applied to the high dimensional dataset of multiple variables. This clustering step aims to group similar data points into different classes within each terrain type. These clustered labels (within each terrain type) become categorical target labels for the boreholes. Subsequently, 1-D logistic regression is applied to all boreholes in each terrain type, with depth as the sole independent variable and the borehole's clustered labels as the target. The 1-D logistic regression model learns the relation between the clustered labels and the depth variable. The result is a stratified borehole profile summarizing each of the four terrain types, as desired.

The four terrain types in the Subedi et al. [1] dataset correspond loosely with the six quaternary surficial geology classes in the Surficial Geology, Lac de Gras map (Ward et al. 2014) [2], as shown in table 5.1.

The Lac de Gras dataset incorporates six ground ice properties, each expressed in percentage format.

- excess ice content

Terrain type (Subedi)	Definition	Heuristic correspondance with Surficial Geology Lac de Gras map
Till	Upland till, smoothly rounded hills, thick till	Hummocky till(Th)
Valley	Fluvially reworked till	Till blanket(Tb), Till veneer(Tv)
Organics	Ice-wedge polygon in peatlands with hummocky surface topography	Organic deposits (O)
Esker	Hill top with sparse vegetation or exposed soil	Esker sediments(GFr), Subaerial outwash fan sediments(GFf)

Table 5.1: Correspondence between four terrain types (Subedi et al.) [1] and classifications in the Lac de Gras surficial geology map [2].

- gravimetric water content
- frozennness
- gravel content
- sand content
- fine particles

Meanshift clustering is applied to borehole data of each of the four terrain types to convert 6-dimensional float value to 1-d categorical value, as seen in Figure 5.5. Higher-dimensional data is visualized in 2-D embedded space through the Python "Uniform Manifold Approximation and Projection (UMAP) for Dimension Reduction" library [24]. The resulting cluster class, encompassing information condensed from 6 dimensions, will represent each sample in the next step.

The next step applies 1-D logistic regression with depth has the sole independent variable, and the cluster class as the target. This process partitions each summary borehole profile into depth intervals, where ground ice properties within each depth interval are uniform.

Regarding the desired format of the summary depth profiles, each depth interval within each terrain type corresponds to a uniform material type (categorical), a triangular distribution (with minimum, maximum, and mode values), as well as the mean and standard deviation of the following:

- gravel
- excess ice
- gravimetric water content
- frozenness

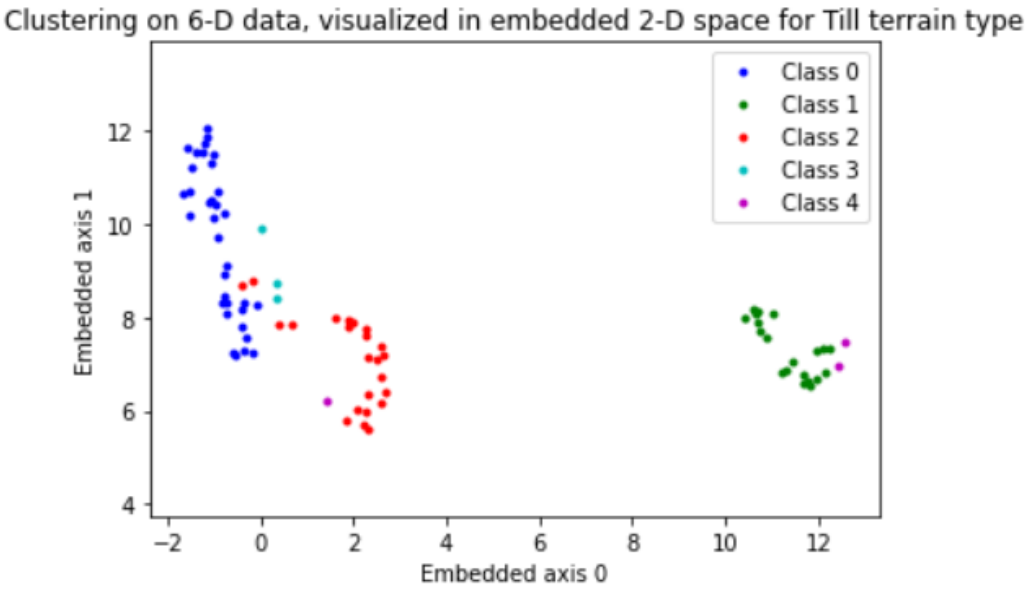


Figure 5.5: Clustering method on the 6-D Lac de Gras region borehole data, visualized in 2-D embedded space. The two embedded axis are for visualization purposes only, and have no significance in real life. The cluster class, a 1-D categorical variable, encompasses information from all six dimensions.

- each of the material types

The resulting depth profiles corresponding to the four terrain types are viewable in the code respository .2.

5.2.3 Method summary

The method to produced summarized borehole profiles for respective terrain types consist of two steps:

- Apply mean-shift clustering on the multi-variable borehole data (the six ground ice properties in this case) to obtain cluster class.
- Apply logistic regression with depth as the independent variable and cluster class as the target.

The borehole profiles of the four terrain types can be utilized in conjunction with the surficial geology map. For instance, figure 5.6 illustrates the hummocky till area as classified by the surficial geology map [2]. Sub-surface characteristics within these regions can be considered part of a cohesive unit, whose sub-surface characteristics are expressed by the depth profile of the terrain type "Till" among the four terrain types. This facilitates exploration in tandem with other surface data in the highlighted zones, such as geomorphometric or remote sensing data, to unveil potential correlations. This avenue opens doors for future

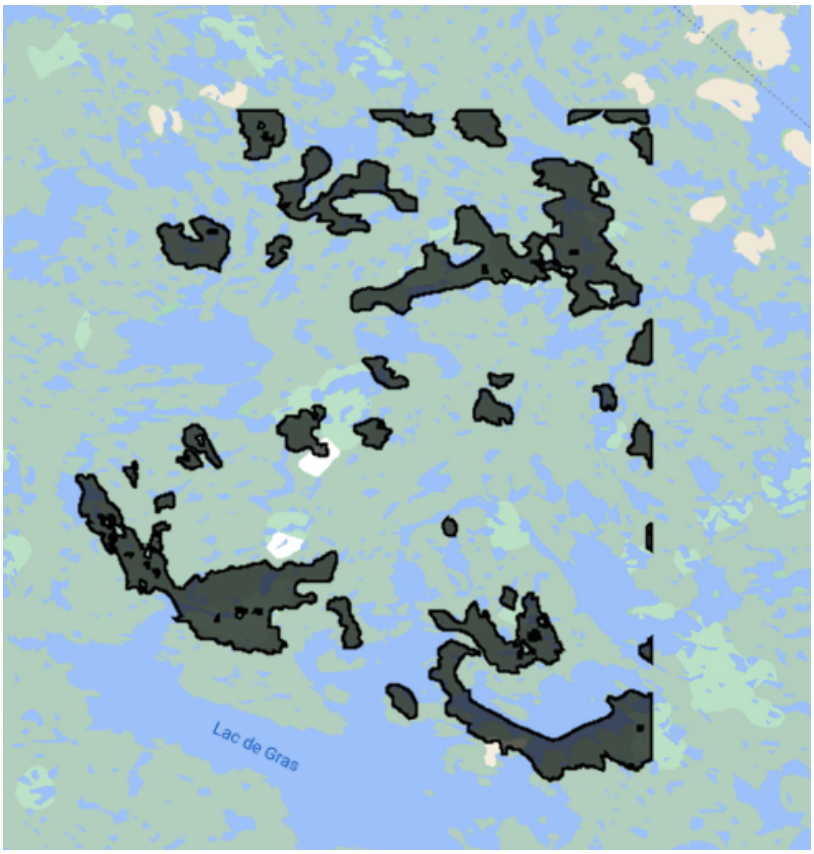


Figure 5.6: Areas classified as "hummocky till" by surficial geology map of Lac de Gras region, visualized on GEE platform.

investigations, enabling analyses on a more macro scale that more closely reflects the scale of state-of-the-art ground-ice research such as GIMC [3].

5.3 Conclusion

The aim of this thesis was to investigate the potential of mapping 3-D ground ice properties.

The implementation of a robust data fusion scheme, integrating multi-format and multi-modal datasets consisting of tabular borehole data and land surface raster data, provides the foundation for mapping ground-ice. Borehole and raster data in the ITH area in Northwest Terrotories, Canada were pre-processed and standardized for deep-learning training.

Applying the custom FiLM framework on an deep learning architecture effectively incorporates both data modalities, The custom FilM model has been verified on modified CIFAR10 test data to show that adding image data that contains useful information through the side branch improves prediction accuracy as expected.

Accuracy scores are compared among XGBoost decision tree, multi-layer perceptron, and FilMed deep learning architectures. Unfortunately, the incorporation of surface data

does not appear to improve accuracy of sub-surface ground ice predictions. A statistical comparison between sub-surface and surface data explores factors contributing to the absence of correlation, such as mismatch in resolution scale and bias in borehole data.

Key insights from this thesis reveal which surface variables may be the most indicative of sub-surface properties, and recommends that, for future efforts in mapping 3-D ground ice properties, it may be beneficial to shift towards more macro scales in the longitude-latitude plane on par with that of the Ground Ice Map of Canada.

These contributions collectively advance the understanding and modelling of ground ice characteristics, paving the way for more nuanced and informed ground-ice research in the future.

Bibliography

[1] R. Subedi, S. V. Kokelj, and S. Gruber. Ground ice, organic carbon and soluble cations in tundra permafrost soils and sediments near a laurentide ice divide in the slave geological province, northwest territories, canada. *The Cryosphere*, 14(12):4341–4364, 2020.

[2] B. C. Ward, L. A. Dredge, and Daniel Ernest Kerr. Surficial geology, lac de gras, northwest territories, nts 76-d, 2014. Includes bibliographical references. Includes abstract in French.

[3] H. B. O'Neill, S. A. Wolfe, and C. Duchesne. New ground ice maps for canada using a paleogeographic modelling approach. *The Cryosphere*, 13(3):753–773, 2019.

[4] Dylan Beaudette, Pierre Roudier, and Jay Skovlin. *Probabilistic Representation of Genetic Soil Horizons*, pages 281–293. 04 2016.

[5] D.E. Beaudette, P. Roudier, and A.T. O'Geen. Algorithms for quantitative pedology: A toolkit for soil scientists. *Computers Geosciences*, 52:258–268, 2013.

[6] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. *CoRR*, abs/1512.03385, 2015.

[7] Marius-Constantin Popescu, Valentina Balas, Liliana Perescu-Popescu, and Nikos Mastorakis. Multilayer perceptron and neural networks. *WSEAS Transactions on Circuits and Systems*, 8, 07 2009.

[8] Xie Jian, Zhang Yunquan, and Qi Yue. Remote sensing image classification based on different convolutional neural network models. In *2021 6th International Symposium on Computer and Information Processing Technology (ISCIPT)*, pages 312–316, 2021.

[9] Danfeng Hong, Qinghua Hu, and Weidong Xiao. Deep residual learning for remote sensing data classification. In *2017 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, pages 4099–4102. IEEE, 2017.

[10] Radhakrishna Achanta, Sabu Emmanuel, P. M. Shankar, Sriram Ganapathy, and Anil K. Sao. Multimodal neural networks: A survey on recent innovations and trends. *Neural Computing and Applications*, 28(11):3151–3169, Nov 2017.

[11] Ethan Perez, Florian Strub, Harm de Vries, Vincent Dumoulin, and Aaron Courville. Film: Visual reasoning with a general conditioning layer. 2017.

[12] PermafrostNet. Permafrostnet erddap: Easier access to scientific data. `https://data.permafrostnet.ca/erddap/index.html`, 2020.

[13] Giuseppe Amatulli, Daniel McInerney, Tushar Sethi, Peter Strobl, and Sami Domisch. Geomorpho90m, empirical evaluation and accuracy assessment of global high-resolution geomorphometric layers. *Scientific Data*, 7(1), May 2020.

[14] Polar Geospatial Center. Arcticdem, September 2018.

[15] O. Conrad, B. Bechtel, M. Bock, H. Dietrich, E. Fischer, L. Gerlitz, J. Wehberg, V. Wichmann, and J. Böhner. System for automated geoscientific analyses (saga) v. 2.1.4. *Geoscientific Model Development*, 8(7):1991–2007, 2015.

[16] GAMMA contributors. Gamma software, 2022.

[17] GDAL/OGR contributors. GDAL/OGR geospatial data abstraction software library, 2020.

[18] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Kopf, Edward Yang, Zachary DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. Pytorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems 32*, pages 8024–8035. Curran Associates, Inc., 2019.

[19] Tianqi Chen and Carlos Guestrin. Xgboost: A scalable and accurate implementation of gradient boosting. *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*, pages 785–794, 2016.

[20] A. Hyvärinen and E. Oja. Independent component analysis: algorithms and applications. *Neural Networks*, 13(4-5):411–430, jun 2000.

[21] Scikit Learn. Fastica on 2d point clouds. `https://scikit-learn.org/stable/auto_examples/decomposition/plot_ica_vs_pca.html#sphx-glr-auto-examples-decomposition-plot-ica-vs-pca-py`, 2023.

[22] Scott M Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, editors, *Advances in Neural Information Processing Systems 30*, pages 4765–4774. Curran Associates, Inc., 2017.

[23] Avanti Shrikumar, Peyton Greenside, and Anshul Kundaje. Learning important features through propagating activation differences. *arXiv preprint arXiv:1704.02685*, 2017.

[24] Leland McInnes, John Healy, and James Melville. Umap: Uniform manifold approximation and projection for dimension reduction, 2020.

.1 Abbreviations

- DEM: Digital elevation model

- FiLM: Feature-wise linear modulation
- GDAL: Geospatial Data Abstraction Library
- GEE: Google Earth Engine
- GIMC: Ground Ice Map of Canada
- HDS: Homogenous distributed scatterer
- ICA: Independent component analysis
- ITH: Inuvik-Tuktoyaktuk highway
- MLP: Multi-layer perceptron
- PCA: Principal Component Analysis
- SHAP: SHapley Additive exPlanations

.2 Code

All code can be viewed at https://github.com/bqianz/film .